

TASK 2 - EXPLORATORY DATA ANALYSIS

E-Commerce Sales Analysis

Internship: Data Analytics

I selected an e-commerce sales dataset for this task. The main purpose was to understand the data first and then find useful patterns from it. I used Python and a few common data analysis libraries for the work.

1. Aim of the Project

The aim of this project is to study sales data and understand which categories, products and cities have more sales. I also checked the data for missing values, duplicate records and unusual values.

2. Tools Used

Python, Pandas, NumPy, Matplotlib, SciPy and Jupyter Notebook were used. Pandas was mainly used for reading and cleaning the CSV file, while Matplotlib was used for charts.

3. Dataset

The dataset contains 1002 rows before cleaning and 10 columns. The columns include order ID, order date, product, category, quantity, unit price, discount, sales, city and payment method. The dataset included a few missing values and duplicate rows so that the cleaning part could also be demonstrated.

4. Questions I Wanted to Answer

Before starting the analysis, I decided to check a few simple questions: Which category has the highest sales? Which products sell more? Which city has more sales? How do sales change during the year? Which payment method is used most often? Are there any unusual sales values?

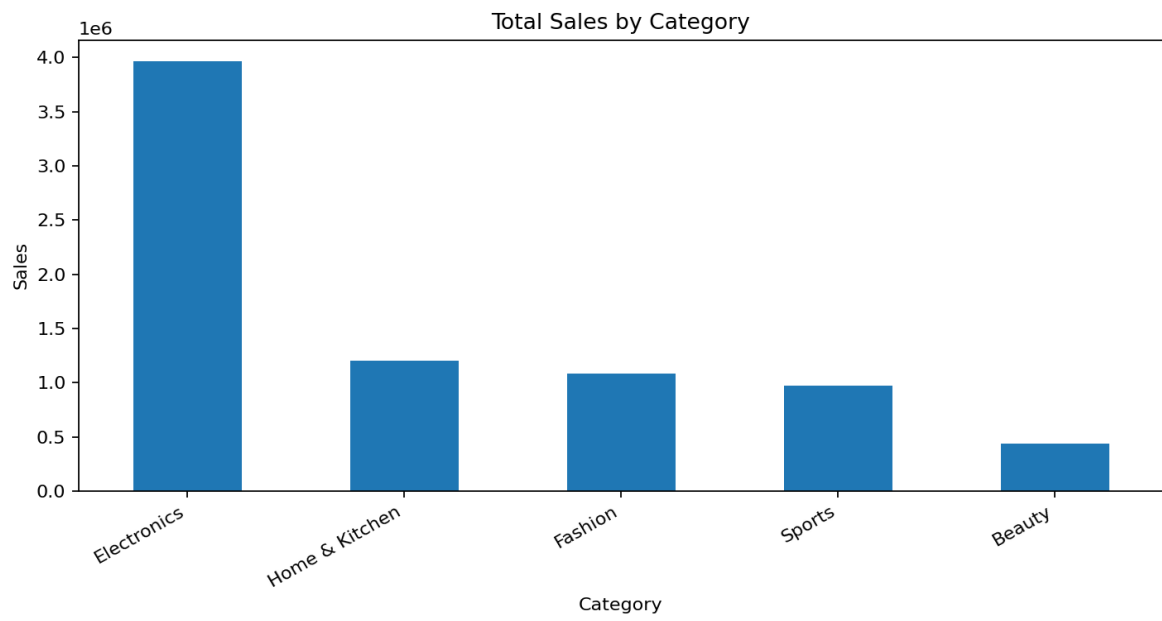
5. Data Cleaning

First, I checked the column names, data types, missing values and duplicate rows. The order date was converted into a date format. Missing city values were filled with the most common city, and missing discount values were filled using the median. Duplicate records were removed before doing the main analysis.

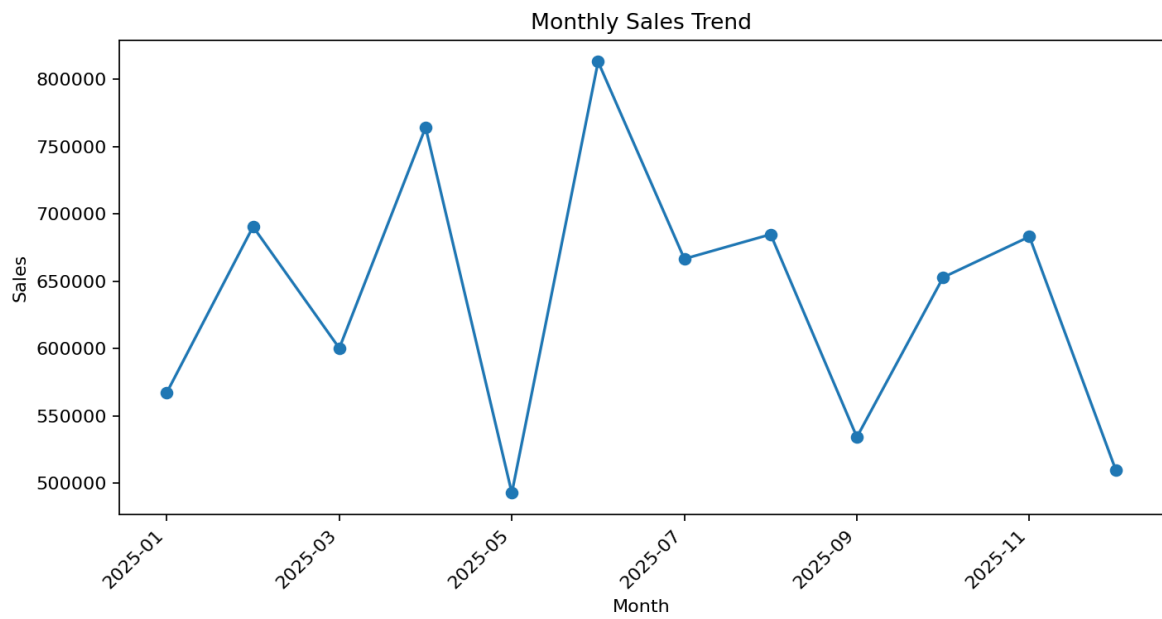
6. Basic Results

After cleaning, the dataset had 1000 rows. Total recorded sales were 7,658,905.22, and the average order value was 7,658.91. The category with the highest total sales was Electronics. The product with the highest total sales was Smart Watch, while Patna had the highest city-level sales. Cash on Delivery was the most frequently used payment method.

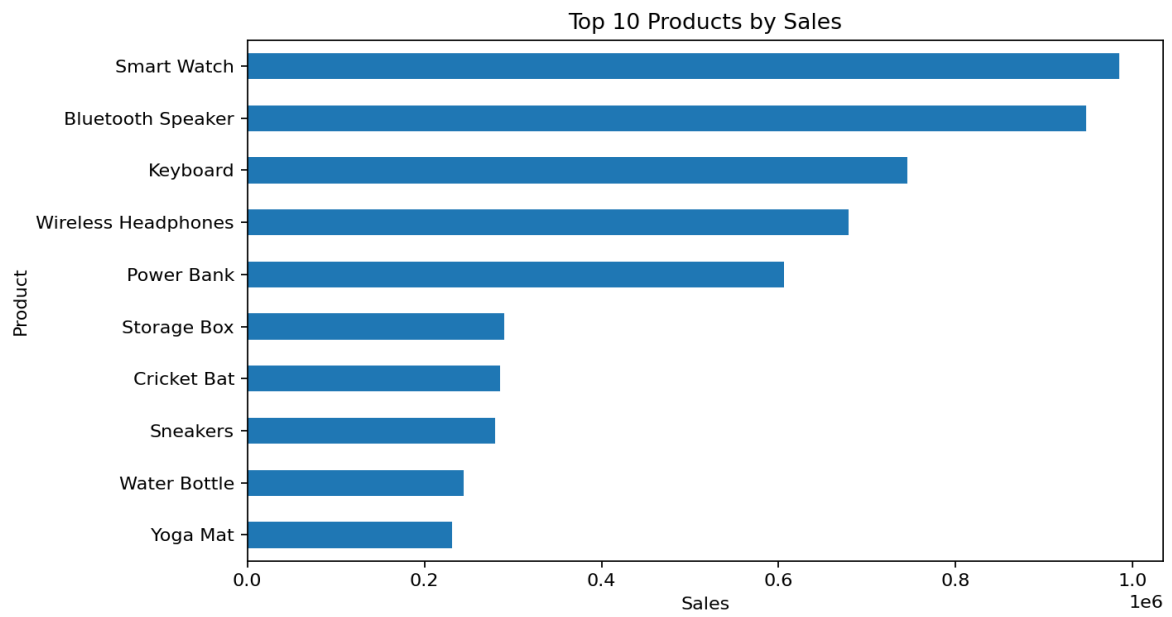
7. Sales by Category



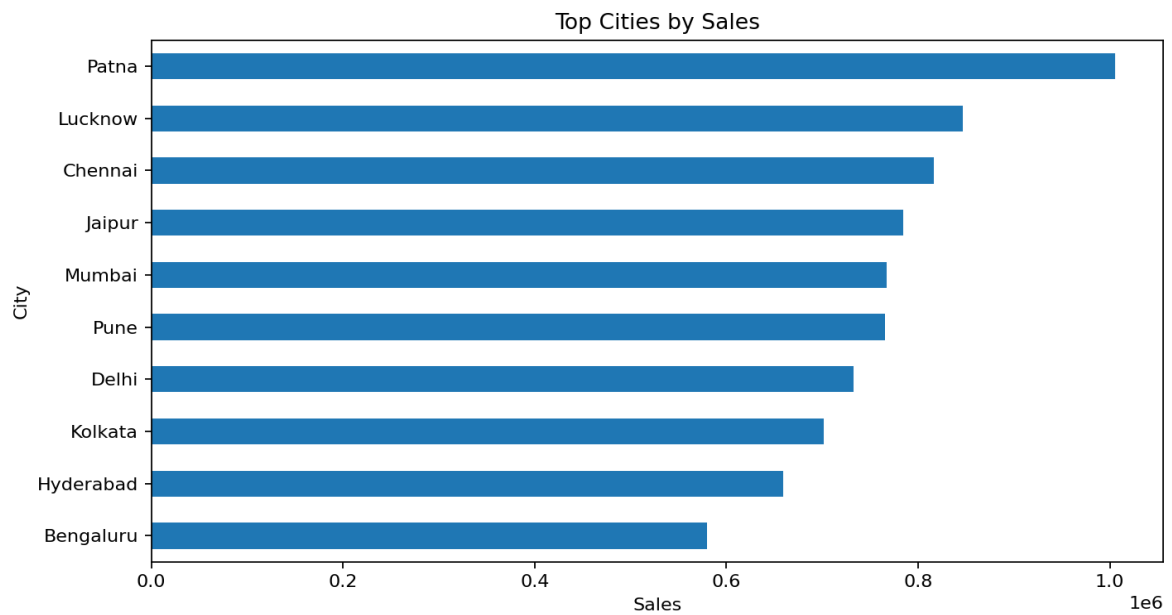
8. Monthly Sales



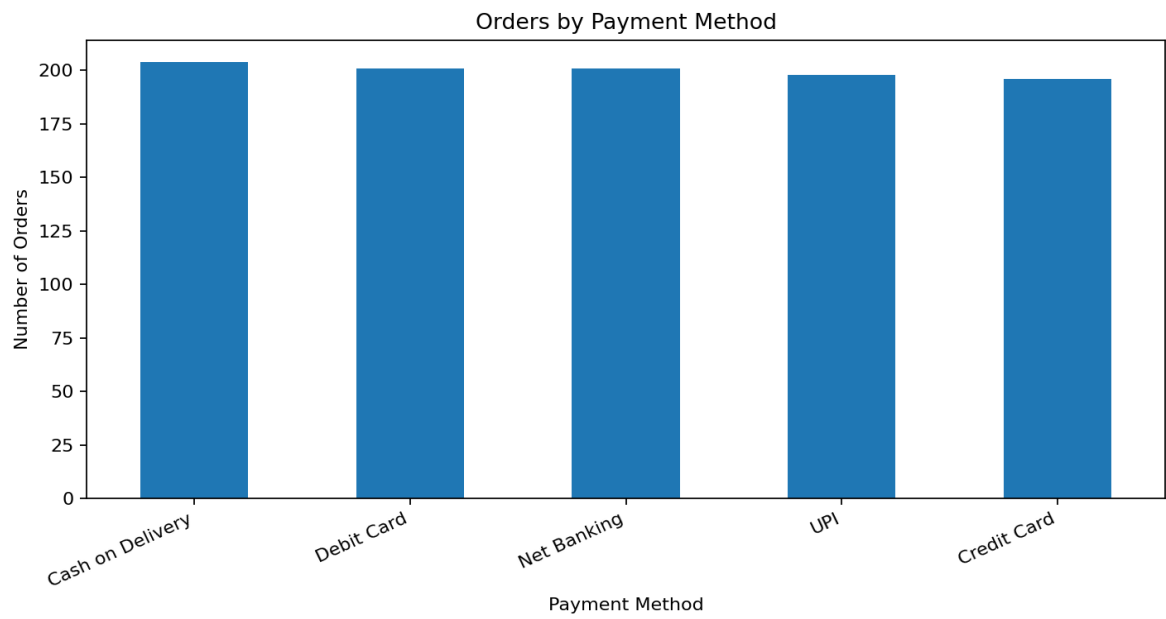
9. Top Products



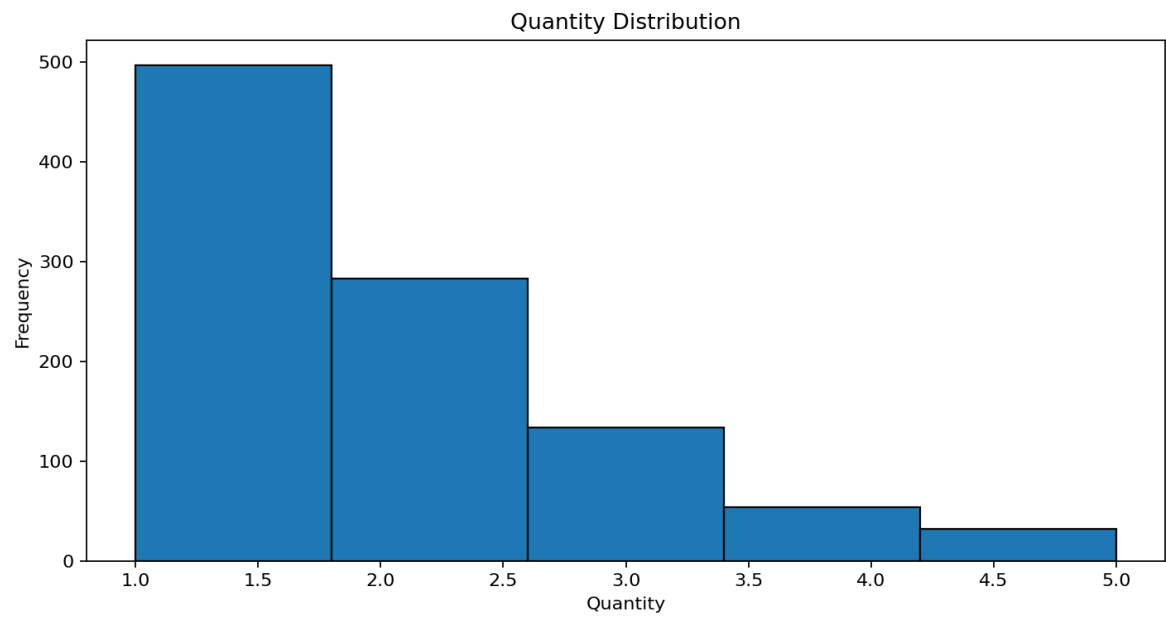
10. Sales by City



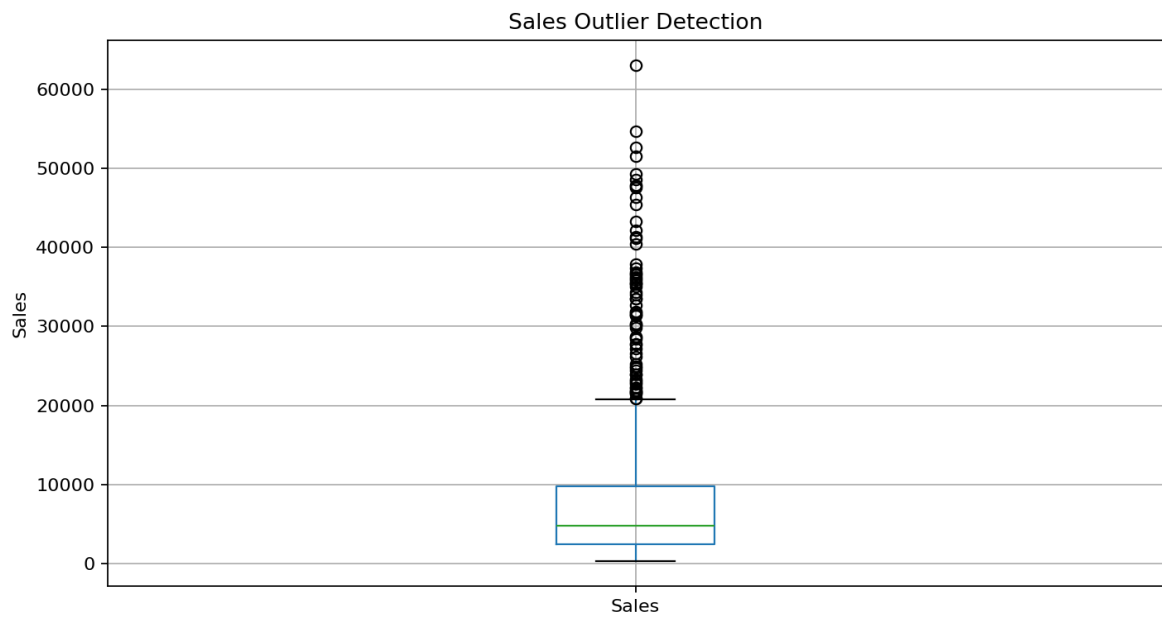
11. Payment Method



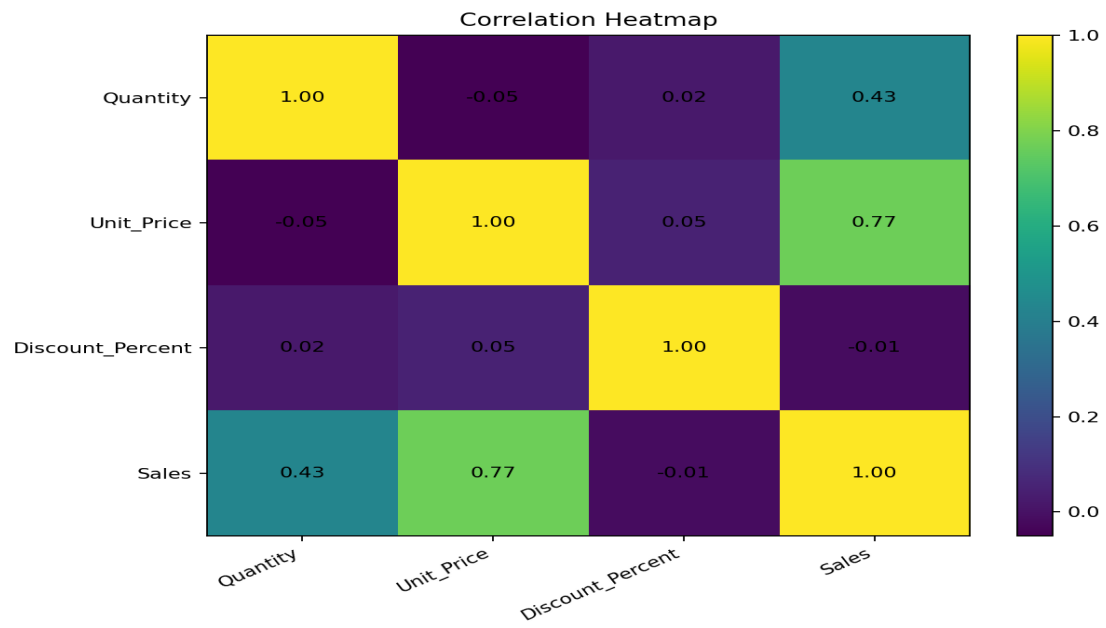
12. Quantity Distribution



13. Checking for Outliers



14. Correlation Between Numerical Columns



15. Hypothesis Test

I also compared the average sales of the different categories using one-way ANOVA. The null hypothesis was that the average sales are the same across the categories. The alternative hypothesis was that at least one category has a different average. The Python notebook contains the F-statistic and p-value produced from the dataset.

16. Outlier Check

Using the IQR method, 69 records were marked as possible sales outliers. An outlier is not automatically an error, so these values should be checked rather than deleted without a reason.

17. What I Learned

While doing this task, I learned how to inspect a CSV file, check data quality, clean missing values, remove duplicate rows, group data and make charts. I also got a basic understanding of correlation, outliers and hypothesis testing.

18. Conclusion

The analysis made the raw sales data easier to understand. The charts helped me compare categories, products, cities and months. The same process can be used on a larger real-world dataset after replacing the demo data with an appropriate public dataset.